Introduction to Gaussian Elimination algorithm

Gaussian Elimination method is a numerical method for solving linear system $Ax = b$, where we assume that A is a square $n \times n$ matrix, x and b are both n dimensional vectors. In the process, the system of equations $Ax = b$ is reduced (by Gaussian elimination) to an upper triangular system $Ux = y$ to be solved through backward substitution.

In the numerical program, vector b is stored as $(n+1)-$th column of matrix A. Let loop k control the elimination step, loop i control $i-$th row accessing and loop j control $j-$th column accessing, the sequential Gaussian Elimination algorithm is described as follows:

Forward Elimination:

for $k = 1$ to $n-1$
  for $i = k+1$ to n
    $a_{ik} = a_{ik}/a_{kk}$;
    for $j = k+1$ to $n+1$
      $a_{ij} = a_{ij} - a_{ik} * a_{kj}$;
    endfor
  endfor
endfor

Note that since the entries in the lower triangular matrix vanish after elimination, their space is used to store multipliers $a_{ik} = a_{ik}/a_{kk}$.

Backward Substitution:

for $i = n$ to 1
  for $j = i+1$ to n
    $x_i = x_i - a_{ij} * x_j$;
  endfor
  $x_i = x_i/a_{ii}$;
endfor

Note that x_i is stored in the space of $a_{i,n+1}$.